

EPFL

FIN-503

Advanced derivatives

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Assignment 1 report

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Introduction

This is the first assignment of the *Advanced Derivatives* course. We present and answer the single question in the next and only section of this report. The code related to our results is joined along this report in our submission.

Exercise 1

We consider a non-dividend-paying stock with current value $S = 100$. In the risk-neutral measure the stock follows a jump-diffusion process

$$\frac{dS}{S} = (r - \lambda^Q \gamma)dt + \sigma dW_t + \gamma dN_t$$

We take $\gamma = -0.08$ (i.e. the stock price is subject to downward 8% jumps), $\lambda^Q = 0.2$ (i.e. on average the stock experiences a jump every 5 years), $\sigma = 0.2$, $r = 0.04$. We compute the value of European call options with expirations $T = 0.02, 0.08, 0.25, 0.5$ (i.e. approximately 1 week, 1 month, 3 months, 6 months), and strikes $K/S = 0.8, 0.9, 1, 1.1$. For every expiration, we then plot the Black-Scholes implied volatility of the option as a function of its strike.

First, we use Merton's closed-form solution of the call price for jump-diffusion processes. It is defined as

$$C(S_0) = E^Q[e^{-rT} \max(S_T - K, 0)] = \sum_{j=0}^{\infty} P(N_T = j) BS_{call}(S_0(1 + \gamma)^j, K, T, \sigma, r, q + \lambda^Q \gamma)$$

We compute the probabilities of $N_t = j$ as long as they are greater than 10^{-12} , and then calculate the call prices for each maturity and strike listed above. Then, we find the Black-Scholes implied volatility of the option for each expiration using Brent's method. Figure 1 shows the Black-Scholes implied volatility under a jump-diffusion model.

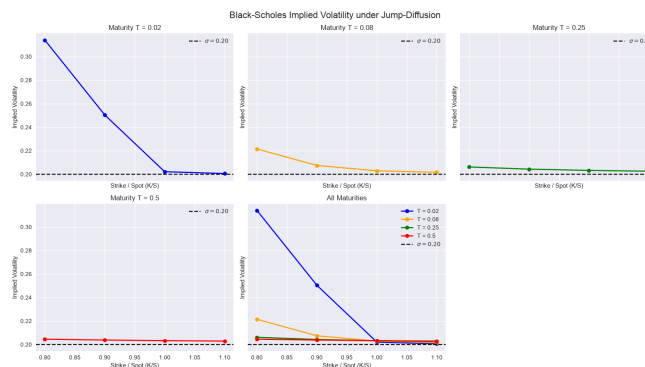


Figure 1: Black-Scholes implied volatility under jump-diffusion for maturities $T = 0.02, 0.08, 0.25, 0.5$ and all of them on the same graph

The implied volatilities show a clear skew. Lower strikes show a much bigger implied volatility, especially for $T = 0.02$ and $T = 0.08$. The implied-volatility curves exhibit a negative skew because the jump component introduces additional downside risk that cannot be captured by a constant-volatility Black-Scholes model. This effect is particularly pronounced for short maturities and low strikes. As maturity increases, the diffusion component becomes relatively more important and the implied-volatility curve becomes flatter, approaching the diffusion volatility of 20%. The downward jump component therefore generates both a strike-dependent implied volatility and a maturity-dependent skew.